

Preoperative Education, Assessment, and Planning Tool

CMT: CMT-0012 (SR)

- Introduction
- Preoperative Planning Criteria Detail
- Risk Factor Assessment
- References
- Footnotes

Introduction

Preoperative planning requires assessment of surgical risk factors, interventions to stabilize or manage underlying problems, and actions to ensure the safety of the patient. Assessment, including medical history, helps to determine the appropriate location for the surgery, intraoperative risks, and postoperative rehabilitation needs. Risk factors for complications may include age, comorbid conditions, and psychosocial issues.(1)(2)(3) Functional classification can be predictive of postoperative morbidity and mortality. Planning includes preoperative education. Education is a multidisciplinary process designed to ready the patient for surgery, decrease morbidity and mortality, and meet the needs of the postoperative recovery course. Clinical judgment is important to identify issues with the preoperative routine (eg, medication management), transition of care, and postoperative management, and should be used throughout the planning process.

This tool is to provide clinicians a general preoperative planning checklist. The purpose is to help identify complicating risk factors and suggest appropriate interventions.

Preoperative Planning Criteria Detail

- **Preoperative routines completed**, as appropriate, which may include(1)(2)(3):
 - Informed consent
 - History and physical examination (preoperative medical evaluation/risk assessment)
 - Medication reconciliation
 - Laboratory, ECG, and imaging studies
 - Autologous blood donation
 - Food and fluid restrictions prior to surgery
 - All allergies (eg, environmental, medication, latex)
 - Insertion of tubes or drains (eg, urinary catheter, nasogastric tube, chest tube), if indicated
 - Body position during surgical procedure
 - Anticipated levels of postoperative pain
 - Cough and deep breathing techniques
 - Mobility and position changes postoperatively
 - Type and screening for blood
- **Anesthesiology interview completed(3)(4)(5)**
 - Existing health concerns and comorbid illnesses and allergies identified and status evaluated, including(2)(6):
 - Dentition status issues identified
 - Cardiovascular status issues identified
 - Hematologic status
 - Personal and family history of blood or bleeding disorders
 - Use of antithrombotic medications
 - Respiratory status, including:
 - Airway assessment
 - History of sleep apnea
 - Tobacco use
 - Neurologic status issues identified
 - Gastrointestinal status issues identified
 - Nutritional status issues identified
 - Renal status issues identified
 - Hepatic function issues identified

- Endocrine function issues identified
- Body mass index above normal limits, see  BMI Calculator to calculate BMI.
- Musculoskeletal status issues identified (eg, kyphosis or scoliosis, temporomandibular joint disorder, cervical or thoracic spine injury)
- Immunologic status issues identified
- Psychiatric issues identified (eg, cognitive status impaired, capacity to consent impaired, substance misuse present)
- Oncology issues identified (eg, currently receiving chemotherapy)
- Preoperative functional capacity impaired
- Perioperative pain management, including(7):
 - Adjustments or continuation of current medications planned
 - Pre-existing pain treated
 - Possible preoperative initiation of therapy for postoperative pain management(2)
 - Optimal use of PCA, epidural or intrathecal opioid analgesia, regional analgesic blocks, or local infiltrations of incisions reviewed
- Preoperative initiation of therapy for postoperative nausea and vomiting management, if appropriate(2)
- Anesthesia type to be used reviewed
- **Postoperative planning initiated(1)(8)(9)**
 - Anticipated discharge destination identified
 - Hospital unit
 - Home
 - Recovery facility
 - Custodial care
 - Other: _____
 - Caregiver identified (and ability confirmed), as appropriate
 - Family member
 - Friend or community member
 - Paid services
 - Other: _____
 - Transportation to next level of care identified
 - Personal vehicle
 - Taxi or hired vehicle
 - Public transportation
 - Other: _____
 - Medical supplies coordinated, as needed (eg, crutches)
- **Preoperative education provided** (verbal and written operative instructions)(A)(1)(2)(3)
 - Patient preoperative care checklist sent and reviewed:
 - Two weeks or more before surgery
 - Day before surgery
 - Day of surgery
 - Whom to call for questions
 - Specific planned surgical procedure to include:
 - Reason for intervention
 - Description of intervention
 - Risks of intervention
 - Preoperative routines specific to location of surgery (eg, informed consent, testing)(9)
 - Preoperative medications(6)
 - Medications to continue (eg, antiarrhythmics, beta-blockers, pulmonary inhalers or nebulized drugs, anticonvulsants, antihypertensives, or psychiatric drugs) taken with a sip of water on morning of surgery
 - Medications to discontinue before surgery; examples include:
 - Aspirin for 3 days preoperatively
 - Clopidogrel for 5 to 7 days preoperatively
 - Nonsteroidal anti-inflammatory drugs (NSAIDs) (eg, ibuprofen, naproxen) for 1 to 3 days preoperatively
 - Naltrexone for 3 days preoperatively(10)
 - Oral hypoglycemic agents withhold the day of surgery(11)
 - Medications to initiate prior to surgery (eg, eye drops)
 - Skin or bowel preparation to be done at home, if indicated(12)
 - Active infections treated
 - Tobacco cessation counseling(13)
 - Support groups to contact before surgery
 - Hospital or surgical center routines and practices reviewed, including:
 - When to arrive at facility
 - Where to park and where to go once patient arrives at facility

- Contact person confirmed
 - Who will bring patient to facility?
 - Who will be available to take patient home?
 - Who will be caregiver the first 24 hours postoperatively?
- What to bring(2)
 - Advance directives, if available
 - Current medication list of prescribed and OTC medication, including vitamins and herbs
 - Insurance cards, Medicare card, and picture identification cards
 - Inhaler
 - BPAP/CPAP personal equipment
- What to leave at home
 - Contact lenses (wear glasses to hospital)
 - Jewelry and other valuables
- What to wear if outpatient surgery being performed
- Anticipated length of stay
- Discharge criteria
- **Postoperative expectations(8)**
 - Destination upon discharge (eg, home, recovery facility)
 - Review of sensations that can occur after surgery (eg, dryness of mouth, sore throat following endotracheal intubation, pain at surgical site)
 - Possible therapy postoperatively (eg, PT, OT, SLP)
 - Potential complications after surgery and how to prevent them(2)(9)
 - Deep breathing and coughing exercises postoperatively to prevent pneumonia and atelectasis
 - Prophylaxis to prevent DVT, including pneumatic compression devices, graduated compression stockings, and pharmaceutical agents(14)
 - Turning exercises to prevent venous stasis, thrombophlebitis, pressure ulcer formation, and respiratory complications
 - Medications for nausea, vomiting, and pain management
 - Early ambulation when ordered to prevent ileus, muscle atrophy, and DVT

Risk Factor Assessment

- **Cardiac conditions** (for noncardiac surgery)(7)(15)
 - Assessment(2)(16)
 - Pacemaker or implantable defibrillator present
 - Drugs or herbal products that may affect coagulation taken (eg, aspirin, ginkgo, ginger)
 - Edema present
 - Neck vein distention present
 - Preoperative testing needed:
 - Exercise stress test
 - ECG
 - Angiography
 - Left ventricular function if history of heart failure
 - Risks identified:
 - Advanced age
 - Diabetes requiring insulin
 - Angina
 - Heart failure
 - Ischemic heart disease
 - High-risk surgery scheduled (eg, vascular, thoracic)
 - Cerebral vascular disease and history of TIA or stroke
 - Renal insufficiency
 - Interventions planned, as indicated
 - Prophylactic beta-blocker therapy before elective surgery for patients with 3 or more risk factors
 - Beta-blockers and ACE inhibitors for systolic dysfunction
 - Fluid balance with heart failure stabilization (eg medications)
 - Percutaneous coronary interventional procedure
 - Bridging instructions for previously prescribed anticoagulant or antiplatelet therapies
 - Tobacco cessation counseling(13)
- **Pulmonary(5)(7)**
 - Assessment
 - Risks identified for postoperative pulmonary complications:
 - Head and neck, thoracic abdominal, neurosurgical, and vascular procedures
 - Emergent procedures

- General anesthesia
- Surgical time over 3 hours
- Prior stroke, cognitive impairment, or impaired sensorium
- Prior intubation
- Functional impairment
- Postoperative nasogastric tube
- COPD
- Obstructive sleep apnea or dyspnea(2)
- Advanced age
- Smoking status (smoking substantially increases risk for wound complications)(13)
- Alcohol consumption of more than 2 drinks per day 2 weeks prior to surgery
- Chronic corticosteroid use
- Preoperative testing needed(9):
 - Chest x-rays
 - Arterial blood gas analysis
 - Pulmonary function testing
 - BUN and albumin levels
- Interventions planned, as indicated
 - Tobacco cessation(13)
 - High-risk COPD patients planned for intensive preoperative treatment with bronchodilators and corticosteroids
 - Preoperative antibiotic treatment of pre-existing infections
 - Lung-expansion maneuvers, deep breathing, and coughing instructions to patient(2)
 - Postoperative exercise plan instructions for patient
- **Alcohol use(7)**
 - Alcohol misuse suspected
 - Interventions planned, as indicated
 - Abstinence encouraged to reduce risks of complications and to shorten recovery time
 - Detoxification interventions
- **Nutrition**
 - Assessment(17)
 - Severe malnutrition (over 15% weight loss within 3 months) present
 - Causes of poor nutrition present:
 - Underlying disease
 - Socioeconomic factors
 - Age
 - Length of hospitalization
 - Anorexia or bulimia
 - Alcohol misuse
 - Interventions planned, as indicated(7)
 - Provide preoperative nutritional support for 7 to 10 days in severely malnourished patients.
 - Dietary regimen followed by patient (eg, low salt with heart failure)
- **Delirium(16)(17)**
 - Assessment
 - Preoperative risks identified(6):
 - Older than 70 years
 - Abnormal serum sodium, potassium, or glucose level
 - Dehydration
 - Dementia
 - Depression
 - Debility or immobility
 - Sensory impairment
 - Functional and cognitive impairment
 - Drug and alcohol misuse
 - Cardiovascular surgery
 - Hip fracture surgery
 - Polypharmacy
 - Postoperative risks identified:
 - Pre-existing cognitive deficit
 - Advanced age
 - Use of meperidine or benzodiazepines, anticholinergics, or antihistamines
 - Use of antihypertensives and beta-blockers
 - Pain

- Postoperative hematocrit under 30% (0.30)
- Urinary catheter use
- Sepsis
- Sensory deprivation and sensory overload
- Restraint use
- Interventions planned, as indicated
 - Early mobilization and walking
 - Orienting communication
 - Therapeutic activities
 - Nonpharmacologic approaches to sleep
 - Adaptive equipment for vision and hearing
 - Maintaining nutrition and hydration
 - Pain control
 - Geriatrician consult
- **Diabetes(11)**
 - Assessment
 - Preoperative testing completed.
 - Blood sugar within acceptable range on day of surgery
 - Diabetes complications present
 - Stroke
 - Myocardial ischemia
 - Infections (ie, wound and urinary tract)
 - Renal failure
 - Interventions planned, as indicated
 - Schedule surgeries and procedures for early morning to lessen effect on patient's treatment plan.
 - Patients on insulin:
 - Insulin is not stopped in type 1 diabetes mellitus.
 - 1/2 of NPH dose or 75-80% of long-acting or pump basal insulin dose on morning of procedure given
 - Patients on oral hypoglycemic agents(18):
 - Hold medication on day of procedure, and resume when tolerating normal diet.
 - Frequently monitor blood sugar before, during, and immediately after procedure.
 - Patients with type 2 diabetes mellitus may require insulin in postoperative period for adequate glycemic control.
 - Possible insulin infusion
 - Postoperative measures planned, including:
 - Rapid transition to preoperative insulin regimen in stable patients (to facilitate timely discharge).
 - Patient monitored closely for:
 - Nausea and vomiting
 - Hypotension
 - Fever
 - Serum potassium levels
 - Proper footwear and heel protection while in bed to prevent foot ulcers in patient with diabetic peripheral neuropathy
 - Potential adverse effects monitored for diabetic autonomic neuropathy, including:
 - Diminished bladder tone with associated risk for urinary retention
 - Gastropathy or gastroparesis aggravated by narcotic analgesics, thus delaying ability for oral nutritional intake
 - Orthostatic hypotension with early postoperative ambulation

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Footnotes

[A] Instructions should take into account the patient's age, comorbidities, health literacy, and type of procedure being performed.(2) [A in Context Link 1]

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Last Update: 3/14/2024 1:13:41 AM
Build Number: 28.1.20240313235330.016020